Topic 13 — Desertification (मरुस्थलीकरण) & Land Degradation (भू-क्षरण)

Covers syllabus (click to expand)

Desertification | Land Degradation | Soil Erosion

Sources baked in: NCERT Geography/Biology, ISRO Desertification Atlas 2016, UNCCD, MoEFCC/NRSA, UPPCS PYQs 2018–2025 **Weight:** ★★ Medium — concept traps, erosion sequence, Chambal ravines, UNCCD **Last verified:** August 2026

Current Affairs (this topic)

- UNCCD / LDN and Indian desertification atlas updates are the CA refresh.
- Bundelkhand / Rajasthan land-degradation colour stays high-yield for UP.

13.1 Desertification

Identity: ASDM drylands | Productivity loss | Overgrazing | UNCCD 1994

- **Desertification** is **land degradation in arid, semi-arid, and dry sub-humid (ASDM) areas** caused by climate variation and human activities.
- It is **not** simply existing deserts like the Thar expanding. Any qualifying dryland that loses fertility and vegetation can desertify.
- Drylands cover about **41% of Earth's land** and support roughly **2 billion people**.
- **Overgrazing** is the **main human cause** in drylands. Livestock remove vegetation, compact soil, and expose topsoil to wind and water erosion.
- **Deforestation**, **fuelwood collection**, and **unsustainable cropping** remove root binding and reduce local moisture retention.
- **Improper irrigation** can cause **salinization** and **waterlogging**, turning fertile irrigated land barren — common in parts of Rajasthan canal commands and the Indo-Gangetic fringe.
- **Drought and climate change (जलवायु परिवर्तन)** kill vegetation and trigger flash erosion on bare soil, creating a feedback loop of further degradation.
- **Wind erosion** dominates true arid zones like the **Thar Desert**, where dunes migrate and bury fields.
- Early desertification can be **reversed** through afforestation, controlled grazing, and water harvesting. Advanced stages may take decades to restore.
- India's **ISRO Desertification Atlas (2016)** mapped **96.4 million hectares (~29.3% of geographical area)** under desertification/land degradation. **Rajasthan** (थार) is the worst-affected state.

> **Logic:** Desertification applies to **dry sub-humid Bundelkhand**, not only sand deserts.

- The **UNCCD (United Nations Convention to Combat Desertification)** was adopted on **17 June 1994** in Paris and entered force in **1996**.
- **17 June** is observed as **World Desertification and Drought Day**.
- **SDG 15.3** targets **Land Degradation Neutrality (LDN) (भूमि क्षरण तटस्थता) by 2030** — avoid new degradation and restore degraded land.

PYQ — UPPCS Prelims 2019, Q42

India aims to achieve land degradation neutrality by:

A. 2025

B. 2030

C. 2035

D. 2040

Show answer

Ans: B — SDG 15.3 target year **2030**.

Concept	Scope
Desertification	ASDM drylands only
Drought	Rainfall deficit event
Land degradation	All land types, all processes

13.2 Land Degradation

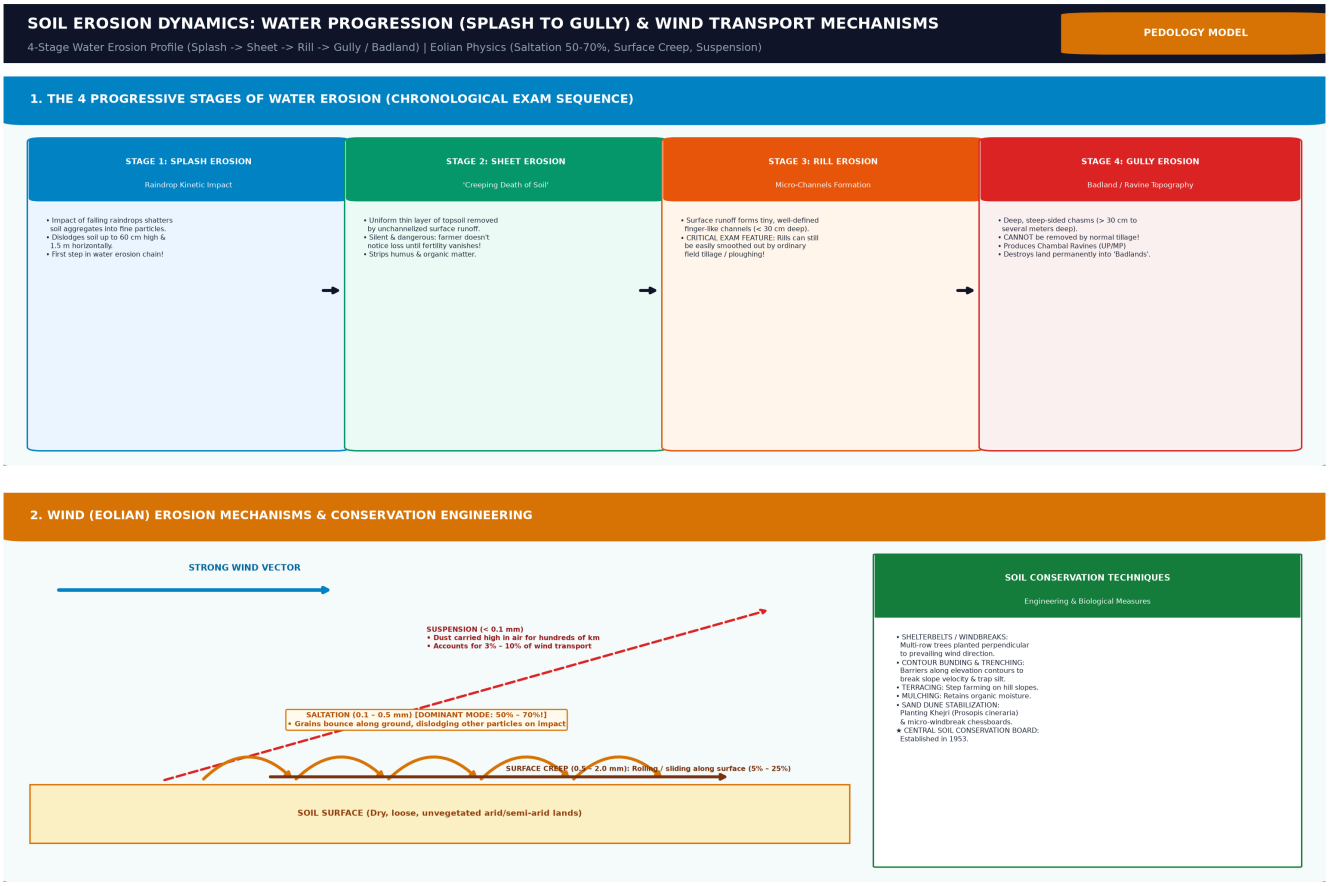
Identity: Productive capacity decline | Erosion + salinity + mining | Wasteland end-stage

- **Land degradation** is the **broader decline in a land's productive capacity** from natural or human factors.
- **Desertification** is one **subset** limited to ASDM drylands. **Soil erosion** is one **major process** within degradation.
- **Human causes dominate** India's degradation statistics: overgrazing, deforestation, unsustainable agriculture, excessive fertilizers, mining, industrial pollution, and urban (नगरीय) sprawl.
- Major degradation **processes** include **soil erosion**, **waterlogging**, **salinization**, **alkalinization**, **mining spoil**, and **deforestation**.
- **Salinization** from excessive canal irrigation affects **Punjab, Haryana, and western UP**.
- **Mining degradation** is severe in **Sonbhadra (UP), Jharkhand, and Odisha**.
- Continued degradation creates **wasteland** — land unsuitable for cultivation. All wasteland is degraded, but not all degraded land is yet wasteland.

- Consequences include **food insecurity**, **rural poverty**, **migration** (Bundelkhand to cities), **reservoir silting**, and **higher flood risk** on silted riverbeds.
- **Salinization and waterlogging** can often be reversed through drainage, gypsum, and drip irrigation. **Deep ravines** are hardest to restore.
- **Afforestation prevents degradation** — it is **not** a cause. Pick it when asked "NOT a cause."
- Policy tools include **Integrated Watershed Management Programme (IWMP)**, **MGNREGA** (contour bunds, check dams), **DPAP**, **Desert Development Programme (DDP)**, and **Soil Health Card Scheme**.

> **Logic:** DPAP (1973)** targets drought-prone arid/semi-arid areas for soil moisture conservation.

Process	Example region
Salinization	Punjab-Haryana canal belt
Mining degradation	Sonbhadra, Jharkhand
Waterlogging	Over-irrigated plains
Ravine formation	Chambal basin



13.3 Soil Erosion

Identity: A-horizon loss | Sheet→ravine | Wind deflation | Conservation measures

- **Topsoil (A-horizon)** is the fertile organic-rich layer, usually **15–30 cm deep**. It can take **centuries to form** but may be lost in **one heavy monsoon** on bare slopes.
- **Normal (geological) erosion** is slow and balanced by soil formation. **Accelerated erosion** from human activity is **10–100× faster** than formation — this is the usual default meaning of "soil erosion."
- **Water erosion** dominates humid and monsoon India. Raindrop impact dislodges soil and runoff carries it downslope.
- **Water erosion sequence:Sheet erosion** (thin uniform layer) → **rill erosion** (क्षुद्र सरिता (रिल) अपरदन) (small channels) → **gully erosion** (deep cuts) → **ravines** (extensive badland networks).
- **Chambal ravines** (चंबल) cover about **3 million hectares** across **MP (Morena, Bhind), UP (Etawah, Agra), and Rajasthan (Dholpur, Karauli)**. They are India's classic **gully/ravine erosion** landscape.
- **Wind erosion** dominates arid zones through **deflation** (lifting fine particles) and **abrasion** (sandblasting surfaces). It affects the **Thar**, Rann of Kutch, and western Rajasthan-UP margins.
- **Gravity erosion** (landslides, slumping) matters in **Himalayan and NE hill states**, especially after deforestation and road cutting.
- **Vegetation cover** is the best natural erosion shield. Deforestation and overgrazing are the top acceleration factors.
- **Conservation measures:contour bunding/ploughing**, **terrace farming**, **shelter belts/wind breaks**, **afforestation**, **gully plugging**, **check dams**, **strip cropping**, and **cover crops**.
- **Earthworms** aerate soil, improve structure, and boost fertility — healthy soil resists erosion.

PYQ — UPPCS Prelims 2025, Q73

1. Earthworms increase soil fertility and are known as friends of farmers.
2. Earthworms have two pairs of hearts.

A. Only 2

B. Neither

C. Both

D. Only 1

Show answer

Ans: D — Statement 1 only. Earthworms have **5 pairs** of hearts, not 2.

Erosion stage	Feature
Sheet	Thin uniform topsoil loss
Rill	Small grooves after rain
Gully	Deep channels, uncrossable by plough
Ravine	Chambal-type badland maze
Agent	Main region

Erosion stage	Feature
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Water	Monsoon slopes, Chambal
Wind	Thar, arid west
Gravity	Himalaya (हिमालय), NE hills